
sshParamiko Documentation

Release 0.1.1

Jonatan Dellagostin

November 16, 2016

1	sshParamiko	1
2	Usage	3
2.1	Executing a simple remote command	3
2.2	Transferring a remote file to a local file	3
2.3	Transferring a local file to a remote file	4
2.4	Installation	4
2.5	Documentation	4
2.6	Source Code	4
2.7	License	4
3	sshParamiko package contents:	5
3.1	sshParamiko package	5
4	Indices and tables	9
	Python Module Index	11

sshParamiko

sshParamiko is a wrapper utility around ssh, using paramiko, to execute commands and exchange files remotely

Usage

2.1 Executing a simple remote command

```
>>> from sshParamiko import RemoteServer
>>> ssh = RemoteServer('/tmp/sshkey', logFolder='.')
>>> ssh.set_log_level('DEBUG')
Log: Changing log level to DEBUG | Log level:DEBUG | Date:01/11/2016 12:04:40
>>> ssh.set_log_rotate_handler(True) # set bziped log files (sshParamiko.debug.log.bz2
# and sshParamiko.error.log.bz2) to be rotated
>>> ssh.connectServer('myServer')
Log: Connecting to server myServer | Log level:DEBUG | Date:01/11/2016 12:04:41
Log: Initiating connection with server myServer... | Log level:DEBUG | Date:01/11/2016 12:04:41
Log: Instantiating transport object for sftp... | Log level:DEBUG | Date:01/11/2016 12:04:41
(True, '')
>>> ssh.executeCmd('whoami')
(True, 'root\n', '')
>>> ssh.closeConnection()
Log: Connection with server myServer ended. | Log level:INFO | Date:01/11/2016 12:04:45
True
```

2.2 Transferring a remote file to a local file

```
>>> from sshParamiko import RemoteServer
>>> ssh = RemoteServer('/tmp/sshkey', logFolder='.')
>>> ssh.set_log_level('DEBUG')
Log: Changing log level to DEBUG | Log level:DEBUG | Date:01/11/2016 12:04:40
>>> ssh.set_log_rotate_handler(True) # set bziped log files (sshParamiko.debug.log.bz2 and
# sshParamiko.error.log.bz2) to be rotated
>>> ssh.connectServer('myServer')
Log: Connecting to server myServer | Log level:DEBUG | Date:01/11/2016 12:04:41
Log: Initiating connection with server myServer... | Log level:DEBUG | Date:01/11/2016 12:04:41
Log: Instantiating transport object for sftp... | Log level:DEBUG | Date:01/11/2016 12:04:41
(True, '')
>>> ssh.getFile('localFile.py', '/root/remoteFile.py', callBack=ssh.transferProgressBar)
Log: Transferring remote file /root/remoteFile.py from server myServer to local file localFile.py
| Log level:DEBUG | Date:01/11/2016 12:08:15
TrueSize: 542 bytes(0.0 MB) || File transfered. [#####] 100.0%
>>> ssh.closeConnection()
Log: Connection with server myServer ended. | Log level:INFO | Date:01/11/2016 12:04:45
True
```

2.3 Transferring a local file to a remote file

```
>>> from sshParamiko import RemoteServer
>>> ssh = RemoteServer('/tmp/sshkey', logFolder='.')
>>> ssh.set_log_level('DEBUG')
Log: Changing log level to DEBUG | Log level:DEBUG | Date:01/11/2016 12:07:40
>>> ssh.set_log_rotate_handler(True) # set bziped log files (sshParamiko.debug.log.bz2 and
# sshParamiko.error.log.bz2) to be rotated
>>> ssh.connectServer('myServer')
Log: Connecting to server myServer | Log level:DEBUG | Date:01/11/2016 12:07:41
Log: Initiating connection with server myServer... | Log level:DEBUG | Date:01/11/2016 12:07:41
Log: Instantiating transport object for sftp... | Log level:DEBUG | Date:01/11/2016 12:07:41
(True, '')
>>> ssh.putFile('localFile.py', '/root/remoteFile.py', callBack=ssh.transferProgressBar)
Log: Transferring local file localFile.py to remote file /root/remoteFile.py in server myServer |
Log level:DEBUG | Date:01/11/2016 12:07:44
TrueSize: 542 bytes(0.0 MB) || File transfered. [#####] 100.0%
>>> ssh.closeConnection()
Log: Connection with server myServer ended. | Log level:INFO | Date:01/11/2016 12:07:44
True
```

2.4 Installation

To install sshParamiko, simply run:

```
$ pip install sshParamiko
```

sshParamiko is compatible with Python 2.6+

2.5 Documentation

<https://sshparamiko.readthedocs.io>

2.6 Source Code

Feel free to fork, evaluate and contribute to this project.

Source: <https://github.com/jonDel/sshParamiko>

2.7 License

GPLv3 licensed.

sshParamiko package contents:

3.1 sshParamiko package

3.1.1 Submodules

3.1.2 sshParamiko.sshParamiko module

class `sshParamiko.sshParamiko.RemoteServer` (*keySSH, **kwargs*)

Bases: `loggers.loggers.Loggers`

Access remote server

Provide a layer of abstraction for accessing, executing commands and transferring files between a host and a server.

Parameters

- **keySSH** (*str*) – path of the ssh private key to connect (must be `None` if using user and password to connect)
- **logFolder** (*str*, **optional**, *default =None*) – folder where the log files of this class will be generated
- **username** (*str*, *optional*, *default =root*) – username using the connection
- **password** (*str*, *optional*, *default =None*) – password for connection if using user and password instead of key
- **sshPort** (*str*, *optional*, *default =22*) – ssh tcp port
- **serverHasDns** (*bool*, *optional*, *default =True*) – if the server is not registered in a DNS domain and/or has not its DNS name equals to its hostname, this flag must be set to `False`, otherwise this condition will be checked to certify we are truly connected to the right server.

closeConnection ()

Description: closes remote server connection

Returns *True* if successfully disconnected, *False* otherwise

Return type `bool`

connectServer (*server*, *ping=True*)

Connects a host and a server via ssh

Parameters

- **server** (*str*) – remote server
- **ping** (*bool*, *default = True*) – if False, ignores if the remote server does not ping back

Returns True if successfully connected, False otherwise

Return type *ret* (*bool*)

Returns Message of error if cannot connect, empty string otherwise

Return type *msg* (*str*)

executeCmd (*cmd*, *timeout=20*)

Executes a command in a remote server shell

Parameters

- **cmd** (*str*) – command
- **timeout** (*str*) – timeout to the command execution (default: 20)

Returns True if command successfully executed, False otherwise

Return type *ret* (*bool*)

Returns command standard output

Return type *output* (*str*)

Returns command standard error

Return type *error* (*str*)

getFile (*localFilePath*, *remoteFilePath*, *callBack=None*)

Transfers a remote file to a local file

Parameters

- **localFilePath** (*str*) – path of the local file
- **remoteFilePath** (*str*) – path of the remote file
- **callBack** (*callback*) – callback that reports file transfer status (bytes transfered and total bytes) Default: None

Returns *True* if successfully transfered, *False* otherwise

Return type *bool*

static pingServer (*server*, *tries=4*)

Connects a host and a server via ssh

Parameters

- **server** (*str*) – remote server
- **ping** (*bool*, *default = True*) – if False, ignores if the remote server does not ping back

Returns True if successfully connected, False otherwise

Return type *ret* (*bool*)

Returns Message of error if cannot connect, empty string otherwise

Return type *msg* (*str*)

putFile (*localFilePath*, *remoteFilePath*, *callBack=None*)

Transfers a local file to a remote file

Parameters

- **localFilePath** (*str*) – path of the local file
- **remoteFilePath** (*str*) – path of the remote file
- **callback** (*callback*) – callback that reports file transfer status (bytes transfered and total bytes) Default: None

Returns *True* if successfully transferred, *False* otherwise

Return type `bool`

static transferProgressBar (*transferredBytes, totalBytes*)

Provides a transfer progress bar

Given a file to be transferred, print a progress bar according to the bytes transferred yet and the file size to be used as a callback for the methods `putFile` and `getFile`.

Parameters

- **transferredBytes** (*str, int or float*) – bytes transferred
- **totalBytes** (*int or float*) – size of file in bytes

validateFiles (*localFilePath, remoteFilePath*)

Checks if a remote and local files has the same sha1sum

Parameters

- **localFilePath** (*str*) – path of the local file to be validated
- **remoteFilePath** (*str*) – path of the remote file to be validated

Returns *True* if files' sha1sums are the same, *False* otherwise

Return type `bool`

3.1.3 Module contents

Indices and tables

- `genindex`
- `modindex`
- `search`

S

`sshParamiko`, [7](#)

`sshParamiko.sshParamiko`, [5](#)

C

`closeConnection()` (ssh-Paramiko.sshParamiko.RemoteServer method), 5
`connectServer()` (sshParamiko.sshParamiko.RemoteServer method), 5

E

`executeCmd()` (sshParamiko.sshParamiko.RemoteServer method), 6

G

`getFile()` (sshParamiko.sshParamiko.RemoteServer method), 6

P

`pingServer()` (sshParamiko.sshParamiko.RemoteServer static method), 6
`putFile()` (sshParamiko.sshParamiko.RemoteServer method), 6

R

`RemoteServer` (class in sshParamiko.sshParamiko), 5

S

`sshParamiko` (module), 7
`sshParamiko.sshParamiko` (module), 5

T

`transferProgressBar()` (ssh-Paramiko.sshParamiko.RemoteServer static method), 7

V

`validateFiles()` (sshParamiko.sshParamiko.RemoteServer method), 7